

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – SCENARIO BASED

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO2 – Manipulation	Assessment:	Assessment Tool 2 – SCENARIO BASED
Weightage:	30% of CIA		
CO5 Statement:	Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)		
Student Name:	CHARANDEEP.N.C	Reg. No.:	192472196
Section / Batch:	CSE-AI	Date:	3-08-2026

Code of Conduct

I Charandeep/192472196 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: *Charandeep*

Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
English Word Classes, Tagsets for English, Part of Speech Tagging	Morphological decomposition of analyzing, analysis, and analytical; identification of roots, affixes, inflectional and derivational forms, and normalization for indexing.	Q1
English Word Classes, Tagsets for English, Stochastic Part of Speech Tagging	Morphological parsing of disagree, agreement, and agreeable; identification of prefixes, suffixes, base forms, and semantic effects of derivation.	Q2
Counting Words, English Word Classes, Part of Speech Tagging	Morphology-based normalization of govern, government, and governance for topic modeling, clustering, and lexical standardization.	Q3
Rule-Based Part of Speech Tagging, Transformation-Based Tagging	Morphological analysis of activate, activation, and reactivation; identification of derivational layers, word-class changes, and semantic shifts.	Q4
Rule-Based Part of Speech Tagging, English Word Classes, Part of Speech Tagging	Inflectional normalization of create, creates, and creating; extraction of root forms and classification of tense/aspect transformations.	Q5

Annexure A – SCENARIO BASED

1. A large-scale search engine indexing system processes user queries containing word variants such as “analyzing”, “analysis”, and “analytical”. The system must improve retrieval accuracy by grouping semantically related terms under a unified representation. Design a morphological processing approach to decompose each input word into root and affixes, distinguish between inflectional and derivational forms, and normalize them for indexing. Apply the process to the given inputs and determine the root form along with the morphological structure of each word.

2. An AI-based sentiment analysis platform receives product reviews containing words like “disagree”, “agreement”, and “agreeable”. The system must interpret sentiment consistently despite variations in word formation. Develop a morphological parsing strategy to identify prefixes, suffixes, and base forms, while capturing how derivational changes influence meaning. Process the given inputs to extract roots and classify each transformation, ensuring accurate semantic interpretation across all word forms.

3. A text analytics pipeline in a news aggregation system handles documents containing terms such as “govern”, “government”, and “governance”. The system must standardize these variations for topic modeling and clustering. Construct a morphology-based normalization mechanism that decomposes each word into its root and affixes, identifies derivational layers, and maps them to a common base form. Apply the approach to the input words and derive the normalized representation along with their morphological breakdown.

4. A document classification system processes input words such as “activate”, “activation”, and “reactivation” to improve semantic grouping and indexing.

Construct a morphological parsing strategy to identify prefixes, suffixes, and root forms while distinguishing derivational layers.

Analyze how prefix addition and suffix transformation affect meaning and word class.

Apply the process to decompose each word and map them to a unified base representation.

Derive the morphological structure and normalized root for each input word.

5. A search optimization module processes input words such as “create”, “creates”, and “creating” to ensure consistent retrieval across grammatical variations. Design a morphology-based normalization approach focusing on inflectional transformations such as tense and aspect.

Identify suffix patterns and map all variations to a common base form without altering meaning.

Apply the process to extract root forms and classify each transformation.

Determine the morphological breakdown and normalized output for each word.

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Q. No. / Criteria Level	Understanding of Scenario	Identification of Key Issues	Application of Concepts	Decision Making	Clarity of Response	Sub. Total	Total %
1							
2							
3							
4							
5							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Scenario 1: Search Engine Indexing – analyzing / analysis / analytical

1. Understanding of the Scenario

A large-scale search indexing system must group semantically related query terms so that a search for one variant retrieves documents containing the others. The words analyzing, analysis, and analytical all relate to the concept of analysis but differ in grammatical form and word class (verb, noun, adjective), which can fragment the index if treated as unrelated terms.

2. Identification of Key Issues

- Multiple surface forms of a single concept exist across different word classes (verb, noun, adjective).
- Inflectional change (progressive aspect marker -ing) must be separated from derivational change (noun/adjective formation).
- A single normalized index key is required to unify retrieval without losing precision.

3. Application of Morphological Concepts

The shared bound root across all three words is 'analy-' (from Greek 'analysis' = to loosen/break up), realized through the base verb 'analyze'.

'Analyzing' is formed by attaching the inflectional suffix '-ing' to the verb 'analyze', marking present-participle/progressive aspect; the word class remains verb-related and meaning is unchanged.

'Analysis' is a derivational noun form built on the same root using the noun-forming suffix '-sis', converting the verb concept into a countable/abstract noun.

'Analytical' is a derivational adjective form built using the adjective-forming suffixes '-tic' and '-al', converting the noun concept into a descriptive adjective.

4. Morphological Decomposition (Decision Output)

Word	Root/Base	Affix(es)	Type	Word Class
analyzing	analyze	-ing (suffix)	Inflectional	Verb (progressive)
analysis	analy-	-sis (suffix)	Derivational	Noun
analytical	analy-	-tic + -al (suffixes)	Derivational	Adjective

5. Decision Making / Rationale

- Index all three forms under the unified base token 'analyze/analysis' so that queries with any variant match the same document cluster.
- Retain the inflectional marker (-ing) only as metadata (aspect tag) rather than a separate index key, since it does not change core meaning.
- Treat the derivational noun and adjective forms as linked synonyms of the same concept node in the search index, rather than merging them into one literal string, to preserve part-of-speech information for ranking.

6. Conclusion (Normalized Representation)

Normalized indexing key: 'analyze / analysis'. All three inputs (analyzing, analysis, analytical) are mapped to this common root cluster, with inflectional information (aspect) and derivational information (word class: verb/noun/adjective) retained as metadata for accurate ranking and retrieval.

Scenario 2: Sentiment Analysis Platform – disagree / agreement / agreeable

1. Understanding of the Scenario

A sentiment analysis platform must interpret product reviews correctly even when reviewers use different derivational forms of the same base concept. The words disagree, agreement, and agreeable share the root 'agree' but differ through prefixation and suffixation, and — critically — the prefix 'dis-' reverses the polarity of sentiment, which must be captured for accurate classification.

2. Identification of Key Issues

- A prefix ('dis-') can invert sentiment polarity, so root-matching alone is insufficient; the affix must be interpreted semantically.
- Suffixes '-ment' and '-able' change word class (verb to noun, verb to adjective) without changing sentiment polarity.
- The system must distinguish meaning-altering derivation (prefix) from class-altering derivation (suffix) to avoid misclassifying 'disagree' as positive.

3. Application of Morphological Concepts

The common root across all three words is the verb 'agree', meaning to hold the same opinion or be in accord.

'Disagree' = prefix 'dis-' (negation/reversal) + root 'agree'. This is derivational because it creates a new word with an opposite meaning (antonym), directly reversing sentiment polarity — the system must tag this as negative sentiment despite sharing the root with positive-sentiment words.

'Agreement' = root 'agree' + derivational suffix '-ment' (noun-forming), converting the verb into an abstract noun referring to the state or result of agreeing; sentiment polarity remains neutral-to-positive.

'Agreeable' = root 'agree' + derivational suffix '-able' (adjective-forming, meaning 'capable of' or 'inclined to'), converting the verb into an adjective describing a pleasant or accommodating quality; sentiment polarity remains positive.

4. Morphological Decomposition (Decision Output)

Word	Prefix	Root	Suffix	Type	Sentiment Effect
disagree	dis- (negation)	agree	—	Derivational	Reverses polarity (negative)
agreement	—	agree	-ment (noun)	Derivational	Neutral/Positive (noun form)
agreeable	—	agree	-able (adj.)	Derivational	Positive (adjective form)

5. Decision Making / Rationale

- Flag prefix 'dis-' as a polarity-inverting morpheme in the sentiment lexicon so 'disagree' is scored as negative even though it shares a root with positive terms.
- Map 'agreement' and 'agreeable' to the same positive-sentiment root cluster as 'agree', since their suffixes only change word class, not polarity.
- Maintain a rule-based affix table (prefix-polarity rules, suffix-class rules) rather than relying on root-matching alone, to ensure accurate sentiment interpretation across all derivational forms.

6. Conclusion (Normalized Representation)

Normalized root: 'agree' (positive polarity). 'Agreement' and 'agreeable' are mapped directly to this root as class-changing derivations; 'disagree' is mapped to the same root but tagged with an inverted polarity due to the negating prefix 'dis-', ensuring correct sentiment classification.

Scenario 3: News Aggregation Topic Modelling – govern / government / governance

1. Understanding of the Scenario

A news aggregation system must cluster articles about governing bodies, institutions, and administrative processes together for topic modelling, even though writers use govern, government, and governance interchangeably. These three forms share a root but represent different derivational layers (verb, concrete/institutional noun, and abstract/process noun).

2. Identification of Key Issues

- Two different nominalizing suffixes ('-ment' and '-ance') produce nouns with subtly different meanings (institution vs. process) from the same verb root.
- Topic models risk treating 'government' and 'governance' as unrelated tokens if only exact string matching is used.
- A normalization layer must preserve the shared conceptual root while optionally retaining the institution-vs-process distinction for finer-grained clustering.

3. Application of Morphological Concepts

The shared root is the verb 'govern', meaning to rule, direct, or control the affairs of a state or organization.

'Government' = root 'govern' + derivational suffix '-ment' (noun-forming), producing a concrete noun referring to the institution or body that governs.

'Governance' = root 'govern' + derivational suffix '-ance' (noun-forming), producing an abstract noun referring to the process, system, or manner of governing.

Both are derivational because they change the word class from verb to noun and add new, distinct shades of meaning, rather than simply marking tense or number.

4. Morphological Decomposition (Decision Output)

Word	Root/Base	Affix	Type	Resulting Meaning
govern	govern	— (base verb)	Base form	Action of ruling/directing
government	govern	-ment (suffix)	Derivational	Institution/body that governs
governance	govern	-ance (suffix)	Derivational	Process/system of governing

5. Decision Making / Rationale

- Normalize all three surface forms to the base lemma 'govern' for coarse-grained topic clustering, so articles using any variant are grouped in the same broad topic.
- Retain a secondary sense tag (institution vs. process) derived from the suffix, allowing finer sub-clustering (e.g., 'government' articles about specific administrations vs. 'governance' articles about policy frameworks).
- Apply a stemming/lemmatization rule that strips '-ment' and '-ance' back to 'govern' during indexing, while preserving the original surface form for display to end users.

6. Conclusion (Normalized Representation)

Normalized representation: root 'govern', with two derivational branches — 'government' (institutional noun) and 'governance' (process noun) — both mapped to the common base 'govern' for topic modelling, while their distinct suffixal meanings are preserved as metadata for precision clustering.

Scenario 4: Document Classification – activate / activation / reactivation

1. Understanding of the Scenario

A document classification system must group documents discussing activate, activation, and reactivation under one semantic cluster. Unlike the previous scenarios, this set introduces both a derivational suffix ('-ion') and a derivational prefix ('re-'), requiring the system to handle layered (nested) morphological structure — a base root with a suffix, further modified by a prefix.

2. Identification of Key Issues

- The word 'reactivation' contains two layers of derivation (prefix + suffix) applied to a single root, requiring layered decomposition rather than single-affix stripping.
- The prefix 're-' adds a repetition/reversal-of-state meaning ('again') rather than simply changing word class, so it must be treated differently from the suffix.
- Word class shifts from verb ('activate') to noun ('activation', 'reactivation'), which affects how the terms should be indexed for classification (action vs. event/state).

3. Application of Morphological Concepts

The core root is 'act' (to do/perform), extended to the derived verb base 'activate' via the verb-forming element '-ate' attached to 'active'.

'Activation' = base verb 'activate' + derivational suffix '-ion' (noun-forming), converting the verb into a noun denoting the act, process, or result of activating.

'Reactivation' = derivational prefix 're-' (meaning 'again') + 'activation'. This is a layered derivation: the innermost layer forms the noun 'activation' from 'activate', and the outer layer adds the repetition prefix, producing a word meaning 'the act of activating again'.

Prefix addition here changes meaning (adds a repetition/aspectual sense) without changing word class further, while the suffix '-ion' is responsible for the verb-to-noun class shift.

4. Morphological Decomposition (Decision Output)

Word	Prefix	Root/Base	Suffix	Layer	Type
activate	—	act(ive)	-ate	Layer 1 (base verb)	Derivational (from 'active')
activation	—	activate	-ion	Layer 2 (verb -> noun)	Derivational
reactivation	re- (again)	activation	—	Layer 3 (prefix on noun)	Derivational (layered)

5. Decision Making / Rationale

- Classify documents using a layered decomposition tree: root 'act' -> 'activate' -> 'activation' -> 'reactivation', so classification can operate at any level of granularity.
- Index 'activate' and 'activation' under the same base concept node, since they differ only by the class-changing suffix '-ion'.
- Keep 'reactivation' linked to the same concept node but flag it with a 'repetition' feature (from the prefix 're-'), which may be relevant for classifying documents about renewals, restarts, or repeated events specifically.

6. Conclusion (Normalized Representation)

Normalized base representation: 'activate' (root concept: act/perform, made active). 'Activation' derives from it via suffixation (verb -> noun), and 'reactivation' derives from 'activation' via prefixation (adding a repetition sense), giving a layered but unified indexing structure: act -> activate -> activation -> reactivation.

Scenario 5: Search Optimization – create / creates / creating

1. Understanding of the Scenario

A search optimization module must retrieve consistent results regardless of the grammatical (tense/aspect) form used in a query. Unlike Scenarios 1-4, the words create, creates, and creating do not involve derivational change at all — they are purely inflectional variants of a single verb, differing only in tense/agreement (creates) and aspect (creating).

2. Identification of Key Issues

- All three forms belong to the same word class (verb) and carry identical core meaning, so over-segmentation into separate concepts must be avoided.
- The system must distinguish inflectional suffixes (grammatical marking) from derivational suffixes (meaning/class change) to avoid unnecessary index fragmentation.
- Retrieval must remain accurate regardless of which grammatical form the user searches with.

3. Application of Morphological Concepts

The root/base form is the verb 'create', meaning to bring something into existence.

'Creates' = root 'create' + inflectional suffix '-s', marking third-person singular present tense agreement; word class and meaning are unchanged.

'Creating' = root 'create' + inflectional suffix '-ing', marking present participle/progressive aspect (an ongoing action); word class remains verb and meaning is unchanged.

Because neither suffix changes word class or core meaning, both are classified strictly as inflectional (not derivational) transformations.

4. Morphological Decomposition (Decision Output)

Word	Root/Base	Suffix	Type	Grammatical Function
create	create	— (base form)	Base	Infinitive/base verb
creates	create	-s	Inflectional	3rd person singular, present tense
creating	create	-ing	Inflectional	Present participle / progressive aspect

5. Decision Making / Rationale

- Apply simple suffix-stripping (stemming) to normalize 'creates' and 'creating' back to 'create', since no derivational/meaning change is involved.
- Do not create separate index entries for each inflected form; instead store tense/aspect as lightweight query metadata for optional result refinement (e.g., filtering by ongoing vs. completed actions).
- Ensure the normalization rule generalizes to other regular verbs with the same inflectional pattern (-s, -ing, -ed) for consistent retrieval across the module.

6. Conclusion (Normalized Representation)

Normalized output: root 'create' for all three inputs. 'Creates' and 'creating' are purely inflectional variants (tense and aspect markers respectively) and are mapped directly to the single base form 'create' for consistent, meaning-preserving retrieval.

Overall Summary Table

The table below consolidates the normalized root/base form and the dominant morphological process identified for each scenario, supporting the clarity and decision-making criteria across the full assignment.

Scenario	Input Words	Normalized Root	Dominant Process
1	analyzing, analysis, analytical	analyze / analysis	Mixed: inflectional + derivational
2	disagree, agreement, agreeable	agree	Derivational (prefix & suffix)
3	govern, government, governance	govern	Derivational (suffix)
4	activate, activation, reactivation	activate	Derivational (layered prefix + suffix)
5	create, creates, creating	create	Inflectional only

Across all five scenarios, distinguishing inflectional morphology (grammatical marking that preserves meaning and word class) from derivational morphology (affixation that changes meaning, polarity, or word class) proved essential for building an accurate, unified normalization layer for indexing, sentiment interpretation, and topic/document classification.